

Enhancing Frequency Stability in Power Systems Using Adaptive Droop-Controlled Battery Energy Storage System

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Abstract—Frequency stability in power systems is increasingly challenged by sudden generation-load imbalances. This paper proposes a novel Hybrid Droop-Adaptive Control strategy with Lead-Lag Washout (LLW) compensation for Battery Energy Storage Systems (BESS) to enhance frequency regulation during load disturbances. The control approach incorporates adaptive tuning of the droop gain, dynamic deadband adjustments, and LLW compensation based on real-time measurements of frequency deviation and Rate-of-Change-of-Frequency (RoCoF). The proposed controller was implemented and validated through dynamic simulations on the standard IEEE 14-bus test system using DigSILENT PowerFactory 2024. The study includes calculation of optimal BESS power and capacity requirements for worst-case load scenarios. Simulation results under different load disturbances including a 20 % sudden load increase and an abrupt bulk load disconnection demonstrate the superiority of the proposed BESS control strategy in improving frequency nadir, reducing stress and ramping demands on synchronous generators, and better preserving spinning reserves. This indicates that our strategy significantly enhances frequency stability during load perturbations. Furthermore, it highlights that BESS effectiveness is contingent upon optimal sizing, making this a critical consideration for planning frequency support systems in modern power networks.

Index Terms—battery energy storage system (BESS), fast frequency response (FFR), droop control, frequency stability, lead-lag washout, DigSILENT PowerFactory

I. INTRODUCTION

The global energy landscape is currently experiencing a profound transformation, primarily driven by the increasing demand for electricity and the accelerated integration of renewable energy sources. As solar, wind, and other variable renewables gain a more substantial share of the energy mix, grid operators face significant challenges in maintaining frequency stability, which is crucial for reliable power system operation. This transition towards cleaner energy sources, while essential

for sustainable development, introduces notable fluctuations between power generation and load consumption, necessitating innovative approaches to ensure grid resilience [1]–[3].

Traditional methods for managing frequency deviations—such as load shedding, dispatching baseload units (e.g., nuclear or coal-fired plants), and utilizing pumped-storage hydroelectricity—have long served as core grid stabilization mechanisms. However, these approaches are increasingly seen as inadequate due to their slow response times and limited operational flexibility, particularly in the face of rapid dynamics introduced by high levels of renewable energy penetration [4]–[6]. The growing share of renewables thus highlights the urgent need for agile, adaptive solutions in modern power systems [7].

In this evolving context, Battery Energy Storage Systems (BESS) have emerged as a pivotal solution. With their high power density and ability to respond within milliseconds, BESS not only balance energy by storing excess generation during low-demand periods but also provide critical ancillary services. Of particular importance is Fast Frequency Response (FFR), where BESS can inject or absorb power instantaneously to counteract frequency deviations [1], [8]. Recent research emphasizes the role of BESS in stabilizing frequency in renewable-rich systems [9].

While conventional droop control remains the standard approach for frequency regulation—modulating power output in response to frequency changes—it relies on fixed droop coefficients. This static nature limits its adaptability to dynamic environments, such as rapidly changing load conditions or fluctuating renewable output. Studies have shown that this rigidity can compromise system stability during disturbances [10], [11]. To overcome this limitation, hybrid approaches that integrate adaptive algorithms with traditional droop control have been proposed [12], [13].

The Hybrid Droop-Adaptive Control strategy with Lead-Lag Washout (LLW) is one such approach, enabling real-time tuning of droop coefficients and deadband parameters combined with dynamic compensation based on system conditions like frequency deviation and rate-of-change-of-frequency (RoCoF). This strategy improves BESS responsiveness to transient events and enhances system reliability, particularly in scenarios where traditional control fails to provide adequate support [14].

The adaptive mechanisms embedded in this strategy continuously monitor grid dynamics and fine-tune controller parameters to optimize BESS response. Previous studies underscore that adaptive droop control not only enhances transient performance but also facilitates fair power sharing among distributed energy resources (DERs), improves energy management, and reduces distribution losses [15], [16]. Furthermore, in interconnected grid environments, such as regional or cross-border power systems, the advantages of adaptive control extend to inter-area frequency regulation and tie-line power balancing. Research shows that such strategies are well-suited to manage complex dynamics in systems with varying inertia and asynchronous operations [17], [18].

This paper investigates the application of an adaptive droop control strategy with LLW for BESS to enhance frequency stability in power systems. To validate the proposed control strategy, this study begins with implementation on the IEEE 14-bus test system to benchmark its performance under controlled, repeatable conditions.

The key contributions of this paper are as follows:

- A novel Hybrid Droop-Adaptive Control strategy with LLW compensation is developed and implemented for BESS to provide dynamic frequency regulation under varying grid conditions.
- The strategy combines adaptive droop gain, deadband control, and LLW compensation, enabling real-time response to disturbances based on system frequency and RoCoF measurements.
- A benchmark study is conducted using the IEEE 14-bus test system to evaluate control performance under standardized conditions.
- Our strategy is validated through time-domain simulations, showing significant improvements in frequency nadir, ramping mitigation, and reserve utilization.

The remainder of this paper is organized as follows. Section 2 details the dynamic modeling of the IEEE 14-bus test system in DigSILENT PowerFactory 2024 and the BESS integration framework. Section 3 presents the hybrid control design, combining adaptive droop gain with dynamic deadband tuning and LLW compensation based on RoCoF. Section 4 investigates the performance of the controller, and discusses the results. Finally, the Conclusion summarizes the findings of our work.

II. POWER SYSTEM NETWORK MODELING USING DIGSILENT POWERFACTORY

The IEEE 14-bus test system was modeled and simulated using DigSILENT PowerFactory 2024 to evaluate the impact

TABLE I: Line parameters of the IEEE 14-bus system used in the simulation study.

Line Name	From Bus	To Bus	R (Ω)	X (Ω)
Line_6-12	Bus_6	Bus_12	1.34	2.79
Line_6-11	Bus_11	Bus_6	1.03	2.17
Line_12-13	Bus_13	Bus_12	2.41	2.18
Line_6-13	Bus_13	Bus_6	0.72	1.42
Line_1-5	Bus_1	Bus_5	9.41	38.86
Line_2-5	Bus_2	Bus_5	9.92	30.30
Line_2-1	Bus_1	Bus_2	6.75	20.62
Line_1-2	Bus_1	Bus_2	6.75	20.62
Line_10-11	Bus_10	Bus_11	0.89	2.09
Line_4-5	Bus_5	Bus_4	2.33	7.34
Line_3-4	Bus_3	Bus_4	11.68	29.80
Line_2-4	Bus_2	Bus_4	10.13	30.72
Line_2-3	Bus_2	Bus_3	8.19	34.49
Line_9-10	Bus_9	Bus_10	0.35	0.92
Line_9-14	Bus_14	Bus_9	1.39	2.94
Line_13-14	Bus_13	Bus_14	1.86	3.79

TABLE II: Generator parameters used in the IEEE 14-bus system.

Name	Bus	P (MW)	Q (MVar)	S (MVA)	V(pu)	Pf
Gen_1	Bus_1	145.0	-15.0	145.8	1.020	0.995
Gen_3	Bus_3	20.0	15.0	25.0	1.010	0.800
Gen_6	Bus_6	15.0	2	15.13	1.020	0.99
Gen_8	Bus_8	15.0	2.0	15.13	1.020	0.99
PV_1	POC_1	10.0	3.3	10.53	0.98	0.95
PV_2	POC_2	25.0	5.1	25.53	1.02	0.98

of BESS FFR on frequency stability. This well-established benchmark network includes multiple synchronous generators, transformers, loads, shunt compensators, and transmission lines, making it ideal for dynamic studies.

The model was developed by replicating the topological and electrical parameters of the original IEEE 14-bus configuration. The BESS was integrated at Bus_14 with a control scheme capable of injecting active power in response to frequency deviations. The BESS power and capacity requirements were determined based on the worst-case load disturbance (20 % sudden increase) and the frequency response requirements of the system. A sizing approach based on the energy needed to maintain frequency deviation within allowable limits (e.g., 49.2 Hz) was adopted, following methodologies outlined in [12] and [19].

The complete network configuration, including line data, generator characteristics, load profiles, shunt compensator, and transformer specifications, is summarized in Tables I to V. The layout of the modified IEEE 14-bus system, including BESS placement at Bus_14 and shunt compensator at Bus_9, is shown in Fig. 1.

III. BESS FFR CONTROLLER DESIGN WITH LLW

A. Frequency stability standards and operational constraints

When generation and demand are balanced, a synchronous generator rotates at a constant speed, maintaining a stable system frequency. However, disturbances such as the sudden disconnection of a generator or abrupt load increases can disrupt this balance, making it more challenging for the power

TABLE III: Load parameters used in the IEEE 14-bus system.

Name	Connection	P (MW)	Q (MVar)	S (MVA)	Pf
Load_10	Bus_10	9.0	5.8	10.7	0.841
Load_11	Bus_11	3.5	1.8	3.9	0.889
Load_12	Bus_12	6.1	1.6	6.3	0.967
Load_13	Bus_13	13.5	5.8	14.7	0.919
Load_14	Bus_14	14.9	5.0	15.7	0.948
Load_2	Bus_2	21.7	12.7	25.1	0.863
Load_3	Bus_3	94.2	19.0	96.1	0.980
Load_4	Bus_4	47.8	-3.9	48.0	0.997
Load_5	Bus_5	7.6	1.6	7.8	0.979
Load_6	Bus_6	11.2	7.5	13.5	0.831
Load_9	Bus_9	29.5	16.6	33.8	0.871

TABLE IV: Shunt compensator data at Bus_9 in the IEEE 14-bus system.

Name	Bus	V (kV)	Q _{max} (MVar)	Q _{act} (MVar)
Shnt_9	Bus_9	33.0	19.0	19.0

system to maintain frequency stability. Such imbalances result in a frequency drop due to insufficient active power generation [20]. Under normal operating conditions, frequency deviations should be limited to within ± 0.2 Hz of the nominal value. In the event of a sudden generation loss or sharp increase in load, the system frequency must not fall below 49.2 Hz and should be restored to at least 49.5 Hz within one minute [19].

B. BESS FFR Controller Design with LLW

The BESS FFR controller with LLW compensation is designed to deliver rapid and optimized support during grid frequency disturbances. As shown in Fig. 2, the measured system frequency f_{meas} is first converted from Hz to per-unit (pu) using the nominal base frequency f_{base} (typically 50 Hz):

$$f_{\text{pu}} = \frac{f_{\text{meas}}}{f_{\text{base}}} \quad (1)$$

The controller then calculates the frequency deviation by subtracting the per-unit frequency from the reference frequency f_{ref} :

$$\Delta f = f_{\text{ref}} - f_{\text{pu}} \quad (2)$$

This deviation signal is processed through the LLW compensator, which provides both transient boost and steady-state droop characteristics. The complete controller structure is mathematically described as follows:

TABLE V: Transformer specifications in the modified IEEE 14-bus system.

Name	HV Bus	LV Bus	HV (kV)	LV (kV)	S _r (MVA)
Trf_4_7	Bus_4	Bus_7	132.0	1.0	10
Trf_4_9	Bus_4	Bus_9	132.0	33.0	10
Trf_5_6	Bus_5	Bus_6	132.0	33.0	60
Trf_7_8	Bus_8	Bus_7	11.0	1.0	60
Trf_7_9	Bus_9	Bus_7	33.0	1.0	60
Trf_POC_1	POC_1	Bus_14	33.0	0.69	50
Trf_POC_2	POC_2	Bus_12	33.0	0.69	50

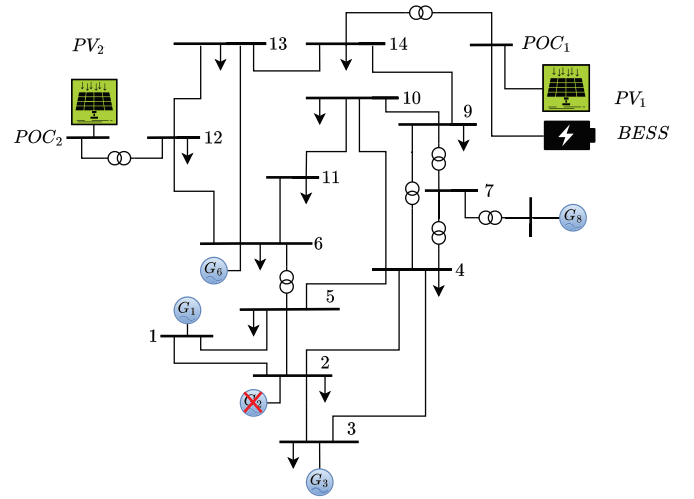


Fig. 1: Modified IEEE 14-bus Transmission network integrated with BESS and PV Plants

$$\Delta f_{\text{pu}}(t) = \frac{f_{\text{ref}} - f_{\text{meas}}(t)}{f_{\text{base}}} \quad (3)$$

$$G_{\text{LL}}(s) = \frac{1 + T_b s}{1 + T_a s} \quad (4)$$

$$y_{\text{LL}}(s) = G_{\text{LL}}(s) \Delta f_{\text{pu}}(s) \quad (5)$$

$$\Delta P_{\text{droop}}(s) = K_d y_{\text{LL}}(s) \quad (6)$$

$$G_{\text{D}}(s) = \frac{s}{1 + T_s} \quad (7)$$

$$G_{\text{W}}(s) = \frac{T_1 s}{1 + T_1 s} \quad (8)$$

$$y_{\text{boost}}(s) = K G_{\text{W}}(s) G_{\text{D}}(s) \Delta f_{\text{pu}}(s) \quad (9)$$

$$\Delta P_{G,\text{raw}}(s) = \Delta P_{\text{droop}}(s) + y_{\text{boost}}(s) \quad (10)$$

$$\Delta P_G(t) = \begin{cases} \Delta P_{G,\text{raw}}(t), & \left| \Delta P_{G,\text{raw}}(t) \right| \leq R_{\text{max}} \\ \text{sign}(\Delta P_{G,\text{raw}}(t)) R_{\text{max}}, & \left| \Delta P_{G,\text{raw}}(t) \right| > R_{\text{max}} \end{cases} \quad (11)$$

$$\Delta P_G(t) = \min(y_{\text{max}}, \max(y_{\text{min}}, \Delta P_G(t))) \quad (12)$$

$$dP_{G1}(t) = \Delta P_G(t) \quad (13)$$

Then, the transfer function of the proposed controller is:

$$\frac{\Delta P_G(s)}{\Delta f_{\text{pu}}(s)} = K_d \frac{1 + T_b s}{1 + T_a s} + K \frac{T_1 s^2}{(1 + T_1 s)(1 + T_s)} \quad (14)$$

Fig. 2 illustrates how the lead-lag block provides phase compensation, the washout filter removes steady-state offset, and the derivative term adds transient boost. The combined transfer function (14) ensures both fast initial response and stable steady-state operation.

TABLE VI: Control parameters of the developed BESS FFR model with LLW implemented in PowerFactory.

Parameter	Value
f_{ref} (Nominal frequency reference)	1.0000 pu
Droop constant	0.0120 pu/Hz
f_{base} (Base frequency)	50.00 Hz
T_a (Lead-lag time constant)	0.8 s
T_b (Lead-lag time constant)	0.2 s
T (Derivative time constant)	0.1 s
T_1 (Washout time constant)	2.0 s
K (Boost gain)	0.3
R_{min} (Ramp rate limit)	-1 pu/s
R_{max} (Ramp rate limit)	+1 pu/s
y_{min} (Minimum output)	-1 pu
y_{max} (Maximum output)	+1 pu

TABLE VII: Technical specifications of the BESS unit used in the FFR study.

Parameter	Value
Connection bus	POC_1
Rated apparent power	1.0 MVA
Power factor	0.95
Phase configuration	3-phase (3PH)
Plant type	Storage
Parallel units	30
Model used	WECC BESS Controller

The filtered frequency deviation signal $y(t)$ is then passed through a deadband function to ignore small deviations within a pre-defined threshold:

$$\Delta f_{db} = \begin{cases} 0, & |y(t)| \leq db \\ y(t), & |y(t)| > db \end{cases} \quad (15)$$

This output signal dP_{G_i} is used as the power reference input (P_{ref}) to the WECC BESS electrical controller (REEC C) model. The controller ensures that the BESS injects or absorbs power in proportion to the frequency deviation with enhanced transient response, thus providing superior frequency stabilization. The LLW compensation improves the ability of the controller to handle both slow and fast frequency variations while maintaining stability.

To model the active power-frequency control behavior of the BESS, we implemented a frequency and voltage controller using the DSL environment of DIgSILENT PowerFactory 2024. The controller uses a droop-based frequency response mechanism with LLW compensation, allowing the BESS to inject active power proportionally to the frequency deviation within a defined deadband while providing enhanced transient response. The key control parameters of the BESS FFR model with LLW compensation—including droop, lead-lag, washout, and limit settings—are listed in Table VI, and were set to ensure rapid and stable dynamic response while preserving coordination with existing synchronous generation assets.

IV. SIMULATION RESULTS AND DISCUSSION

A. Case 1: 20% Sudden Load Increase

Fig. 3 presents the system response to a 20% step increase in load introduced at $t = 10$ s and subsequently restored

to its pre-disturbance level at $t = 50$ s. The comparison is made between scenarios with and without BESS FFR using the proposed LLW controller. As shown in Fig. 3a, the load increase produces a sudden generation-load imbalance, followed by a second transient during load restoration.

The frequency response in Fig. 3b shows that, without BESS FFR, the system experiences a deeper frequency nadir following the load increase and a noticeable frequency excursion during restoration. When LLW-based FFR is enabled, both the initial frequency decline and the subsequent recovery transient are significantly mitigated. The rate of frequency change is reduced, and the system returns to nominal conditions more rapidly, demonstrating improved primary frequency control during both disturbance and recovery phases.

Fig. 3c illustrates the BESS active power response, where rapid power injection occurs immediately after the load increase at $t = 10$ s, followed by a controlled reduction as system balance is restored. Upon load restoration at $t = 50$ s, the BESS responds again by absorbing power to counteract the sudden surplus. The LLW compensation ensures a smooth and well-damped response in both events, avoiding excessive oscillations or sustained power exchange.

As shown in Fig. 3d, photovoltaic generation remains largely unchanged throughout the simulation, indicating that the observed frequency improvements are primarily driven by BESS-based FFR. The fast BESS response preserves spinning reserves by reducing the burden on synchronous generators during both the disturbance and restoration intervals.

Grid losses presented in Fig. 3e are marginally lower in the LLW-enabled case across the entire simulation horizon. This reduction is attributed to smoother power redistribution and reduced transient stress on network components. Fig. 3f further shows that synchronous generators experience sharper ramping in the absence of FFR, whereas LLW-based BESS support moderates generator output during both the load increase and its restoration, thereby reducing mechanical stress and enhancing operational reliability.

B. Case 2: Loss of Bulk Load (Load 3)

Fig. 4 compares system performance following the sudden disconnection of Load 3 at $t = 10$ s, with the load restored at $t = 50$ s. As shown in Fig. 4a, the load shedding event creates an immediate generation surplus, followed by a second transient when the load is reconnected.

The frequency dynamics in Fig. 4b reveal a pronounced frequency overshoot in the non-FFR case following load disconnection, as well as secondary oscillations during load restoration. In contrast, when LLW-based BESS FFR is employed, both the overshoot and subsequent recovery oscillations are substantially damped. The controller effectively limits frequency excursions during both events, demonstrating robust transient performance under opposing disturbance types.

The BESS output power in Fig. 4c shows rapid active power absorption immediately after load shedding, followed by controlled power injection when the load is restored at $t = 50$ s. The LLW compensation enhances the transient

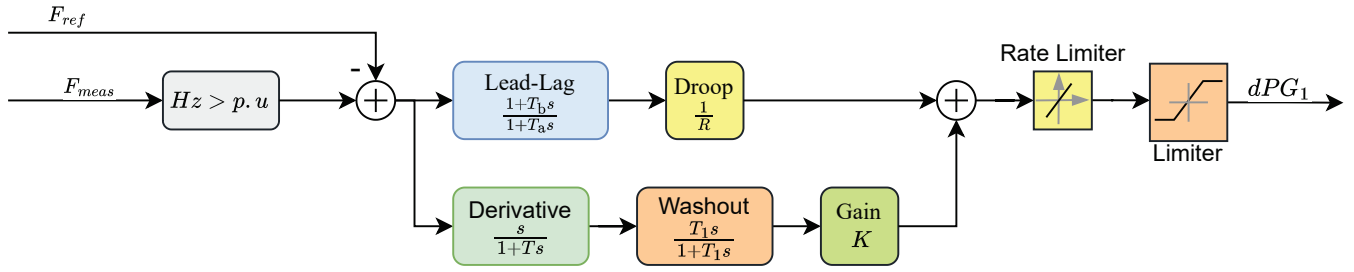
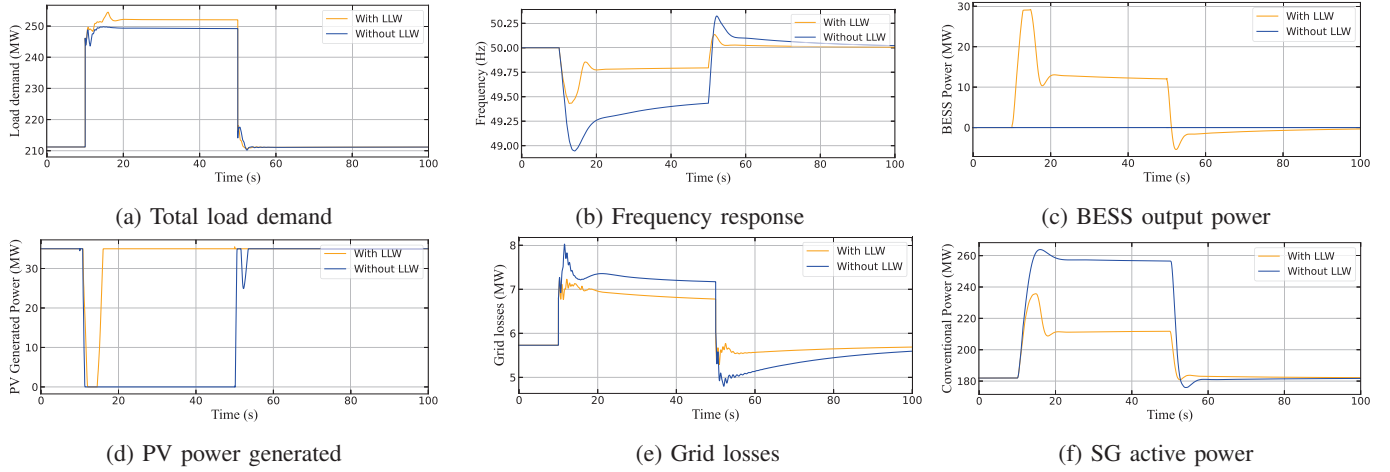


Fig. 2: Block diagram of the BESS FFR controller with LLW compensation

Fig. 3: System performance metrics with and without BESS FFR using LLW during a 20% load disturbance introduced at $t = 10$ s and restored at $t = 50$ s.

response by ensuring fast yet stable power modulation, preventing excessive cycling and maintaining smooth system dynamics.

Fig. 4d confirms that photovoltaic generation remains largely unaffected throughout the disturbance sequence, reinforcing that the observed improvements are attributable to BESS-based FFR. Grid losses shown in Fig. 4e are lower in the LLW-enabled case over the full simulation period, reflecting improved power flow efficiency during transient operation.

Finally, Fig. 4f demonstrates that synchronous generator output is significantly better moderated when LLW-based FFR is applied. Compared to the abrupt output reductions observed without FFR, the BESS-supported case limits generator ramping during both load shedding and restoration, thereby reducing mechanical wear and improving overall system stability.

Overall, the results from both Case 1 and Case 2 confirm that BESS FFR with LLW compensation significantly enhances frequency stability during sudden load changes and their subsequent restoration. By providing rapid and well-damped active power support in both directions, the proposed control strategy improves system resilience, preserves spinning reserves, reduces generator stress, and enhances operational efficiency in low-inertia power systems.

V. CONCLUSION

This study introduced a novel Hybrid Droop-Adaptive Control strategy with Lead-Lag Washout compensation for BESS to enhance frequency stability. Time-domain simulations on the IEEE 14-bus system demonstrated that the proposed controller significantly improves frequency regulation during severe load disturbances. By providing optimized transient support, the strategy effectively reduces the frequency nadir, limits the RoCoF, alleviates ramping stress on synchronous generators, and better preserves spinning reserves.

Future work should focus on validating the performance of the controller in larger, multi-area networks and in systems with low inertia dominated by inverter-based resources. Furthermore, a comprehensive methodology for the optimal sizing and placement of BESS, incorporating economic and reliability constraints, remains to be developed, alongside a comparative analysis against other state-of-the-art adaptive control approaches. In addition, the strategy should be refined to account for real-world practical challenges such as measurement noise and interoperability with existing grid codes. Testing via Hardware-in-the-Loop platforms and developing guidelines for coordination with conventional generation are essential next steps to facilitate the deployment of this advanced BESS control in modern power systems.

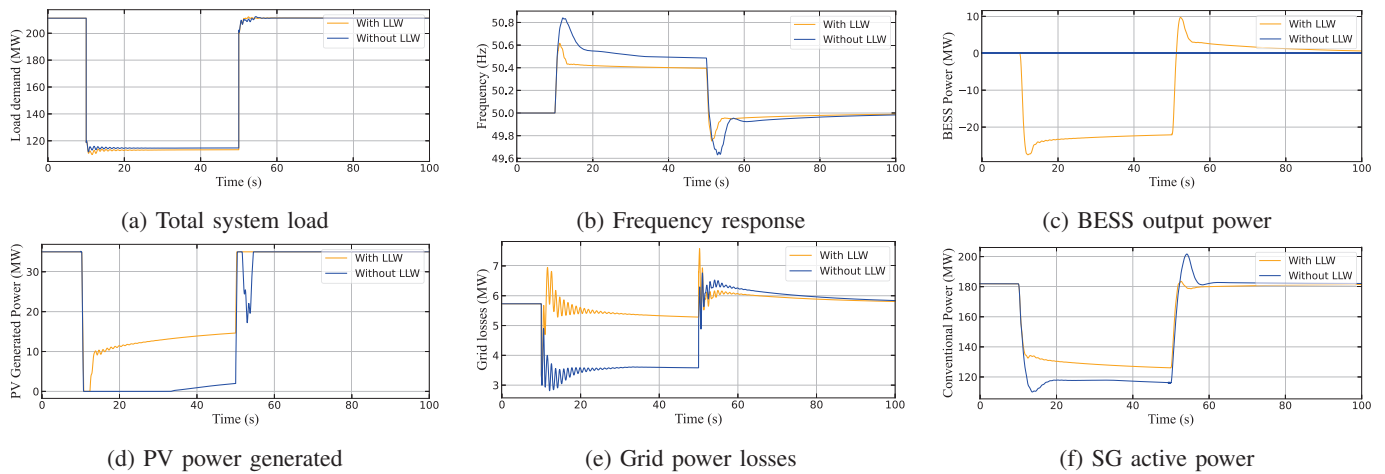


Fig. 4: Dynamic system response with and without BESS FFR using LLW following a bulk load disconnection at $t = 10$ s and restoration at $t = 50$ s.

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